

DATABASE FAILOVER STANDBY STRATEGIES

Cold Standby vs Warm Standby vs Hot Standby
Complete Guide with Layman Explanations

AWS RDS • Aurora • DynamoDB • ElastiCache

Interview & System Design Preparation

TABLE OF CONTENTS

1. Overview & Layman Analogy
2. Cold Standby
3. Warm Standby
4. Hot Standby
5. Side-by-Side Comparison
6. Visual Timeline of Failure
7. AWS Database Failover Options
8. When to Use What
9. Interview-Ready Answers

1. Overview & Layman Analogy

DATABASE FAILOVER — When the primary database goes down, how quickly and safely can a standby database take over? The answer depends on what type of standby you use.

The Backup Driver Analogy

Imagine a bus driver suddenly faints while driving. How quickly can a backup driver take over? That depends on where the backup driver is and how ready they are.

COLD Standby:
Backup driver is AT HOME sleeping
Call him -> wakes up -> gets dressed -> drives to depot
Takes HOURS

WARM Standby:
Backup driver is AT THE DEPOT, sitting in break room
Call him -> walks to bus -> starts driving
Takes MINUTES

HOT Standby:
Backup driver is SITTING IN THE BUS next to main driver
Main driver faints -> backup grabs wheel INSTANTLY
Takes SECONDS

The Basic Setup

```
PRIMARY DATABASE (active - handling all reads/writes)
      |
      | replicates data to...
      v
STANDBY DATABASE (backup - waiting to take over)
```

The difference is HOW READY the standby is.

2. Cold Standby

STANDBY IS OFF — Not running. Recovery relies on restoring from periodic backups. Cheapest option but slowest recovery with potential data loss.

How It Works

+-----+		+-----+
PRIMARY		STANDBY
(ON)	no sync	(OFF)
Running		Shutdown
+-----+		+-----+
- Standby server is NOT running - Periodic BACKUPS taken (daily/hourly) - Backups stored in S3 or file storage - On failure: restore from backup -> start server		

Recovery Steps

1. Detect failure	(minutes)
2. Provision new server	(minutes)
3. Restore from backup	(minutes to hours)
4. Apply transaction logs	(minutes)
5. Redirect traffic	(minutes)
Total Recovery Time: HOURS	
Data Loss: YES - anything since last backup is LOST	

Key Characteristics

Aspect	Detail
Standby State	OFF — not running
Replication	Periodic backups (not continuous)
Recovery Time	Hours
Data Loss	YES — since last backup
Cost	Lowest (no standby server running)
Failover	Manual
AWS Example	RDS Snapshot Restore

4. Hot Standby

STANDBY IS ON AND FULLY SYNCED — Running and receiving data synchronously in real-time. Most expensive but near-instant recovery with zero data loss.

How It Works

```
+-----+ sync replication +-----+
| PRIMARY | =====> | STANDBY |
| (ON)    | (real-time) | (ON)   |
| Active  | (zero lag)  | Ready  |
| Read +  |              | Can take|
| Write   |              | over NOW|
+-----+              +-----+
```

- Standby is running and FULLY in sync
- Data replicated SYNCHRONOUSLY
(every write waits for standby confirmation)
- Standby is SAME SIZE as primary
- On failure: automatic failover, near-instant

Recovery Steps

1. Detect failure (seconds)
2. Automatic failover triggered (seconds)
3. DNS updated (same endpoint) (seconds)

Total Recovery Time: 30-60 SECONDS

Data Loss: ZERO - standby has every transaction

Key Characteristics

Aspect	Detail
Standby State	ON — fully synced in real-time
Replication	Synchronous (zero lag)
Recovery Time	Seconds (30-60s)
Data Loss	ZERO
Cost	Highest (full duplicate running 24/7)
Failover	Automatic
AWS Example	RDS Multi-AZ

AWS Example

RDS Multi-AZ:

Primary (AZ-a) ==sync==> Standby (AZ-b)

- Same endpoint: mydb.xxxx.rds.amazonaws.com
- Failover is AUTOMATIC
- Application doesn't even know it happened

- DNS flips to standby in ~30-60 seconds

5. Side-by-Side Comparison

Feature	Cold	Warm	Hot
Standby state	OFF	ON (idle)	ON (fully synced)
Replication	Periodic backups	Async (slight lag)	Sync (zero lag)
Recovery time	HOURS	MINUTES	SECONDS
Data loss	YES (since backup)	Possible (few sec)	ZERO
Cost	Lowest	Medium	Highest
Failover type	Manual	Semi-automatic	Automatic
Standby size	No server	Can be smaller	Same as primary
AWS example	Snapshot restore	Read Replica	Multi-AZ

Quick Mental Model

Cold = Backup is SLEEPING -> Hours to recover
Warm = Backup is AWAKE but idle -> Minutes to recover
Hot = Backup is READY at wheel -> Seconds to recover

Cost: Cold < Warm < Hot
Speed: Cold < Warm < Hot
Safety: Cold < Warm < Hot

6. Visual Timeline of Failure

The primary database fails at exactly 2:00:00 PM. Here is what happens with each standby type:

Cold Standby Recovery

2:00 PM Failure detected
2:05 PM Team notified, incident created
2:15 PM New server provisioned
2:45 PM Backup restored from last snapshot
3:00 PM Back online

Downtime: ~1 HOUR
Data lost: Everything since last backup

Warm Standby Recovery

2:00:00 Failure detected
2:00:30 Promote replica command issued
2:02:00 Replica promoted to primary
2:03:00 Traffic redirected to new primary

Downtime: ~3 MINUTES
Data lost: Last few seconds (async lag)

Hot Standby Recovery

2:00:00 Failure detected
2:00:05 Automatic failover triggered by AWS
2:00:35 DNS updated, standby is now primary
2:00:40 Traffic flowing to new primary

Downtime: ~30-60 SECONDS
Data lost: ZERO

Side-by-Side Timeline

	2:00	2:05	2:15	2:30	2:45	3:00
COLD:	FAIL	----	notify--	provision-----	restore-	ONLINE
	=====					
	~1 hour downtime					
WARM:	FAIL-promote-	ONLINE				
	=====					
	~3 min					
HOT:	FAIL-ONLINE					
	==					
	~30 sec					

7. AWS Database Failover Options

Amazon RDS

Feature	Type	Recovery	Data Loss
Multi-AZ	Hot Standby	30-60 seconds	Zero
Read Replica	Warm Standby	Minutes	Possible (few sec)
Snapshot Restore	Cold Standby	Hours	Since last snapshot

Amazon Aurora

Feature	Type	Recovery	Data Loss
Aurora Replica	Hot Standby	< 30 seconds	Zero
Aurora Global DB	Warm Standby	< 1 minute	< 1 second lag
Snapshot	Cold Standby	Hours	Since last snapshot

Amazon DynamoDB

Feature	Type	Recovery	Data Loss
Global Tables	Hot (active-active)	Automatic	Zero (multi-region)
Point-in-time Recovery	Cold Standby	Minutes	Restore to any second

Amazon ElastiCache (Redis)

Feature	Type	Recovery	Data Loss
Multi-AZ with auto-failover	Hot Standby	Seconds	Minimal
Backup / Restore	Cold Standby	Minutes	Since last backup

Aurora vs RDS Comparison

Aurora advantages over standard RDS:

- Up to 15 read replicas (RDS has 5)
- Faster failover (< 30 sec vs 60 sec)
- Storage auto-scales (up to 128 TB)
- 6-way replication across 3 AZs
- Backtrack (rewind DB to any point)
- Global Database for cross-region

8. When to Use What

Use Cold Standby When:

- ■ Cost is the top priority
- ■ Application can tolerate hours of downtime
- ■ Dev / staging environments
- ■ Non-critical applications
- ■ Data loss of a few hours is acceptable

Use Warm Standby When:

- ■ Need faster recovery than cold
- ■ Can tolerate a few minutes of downtime
- ■ Can accept losing a few seconds of data
- ■ Want cross-region disaster recovery
- ■ Read-heavy workloads (use replica for reads too)

Use Hot Standby When:

- ■ Zero data loss is critical (banking, healthcare)
- ■ Downtime must be under 1 minute
- ■ Compliance requires synchronous replication
- ■ Mission-critical production systems
- ■ SLA requires 99.99% uptime or higher

Decision Matrix

Priority	Choose	AWS Service
Lowest cost	Cold	RDS Snapshots
Balance cost + speed	Warm	RDS Read Replica
Zero data loss	Hot	RDS Multi-AZ
Zero data loss + global	Hot + Warm	Multi-AZ + Aurora Global DB
Active-active multi-region	Hot	DynamoDB Global Tables

9. Interview-Ready Answers

Q: What is the difference between warm and hot standby?

"Warm standby uses asynchronous replication — the standby is running but has a slight data lag. Recovery takes minutes and you might lose a few seconds of data. Hot standby uses synchronous replication — the standby is fully in sync with zero lag. Recovery takes seconds with zero data loss. In AWS, Read Replicas are warm standby and Multi-AZ is hot standby."

Q: How does RDS Multi-AZ failover work?

"RDS Multi-AZ maintains a synchronous standby replica in a different Availability Zone. When the primary fails, AWS automatically updates the DNS record to point to the standby, which is promoted to the new primary. The application uses the same endpoint, so no code changes are needed. The entire failover takes about 30-60 seconds with zero data loss."

Q: When would you choose warm over hot standby?

"I would choose warm standby when the application can tolerate a few minutes of downtime and a few seconds of data loss, especially for cross-region disaster recovery where synchronous replication would add too much latency. For example, using an Aurora Global Database as a warm standby in another region with less than 1 second of replication lag. Hot standby is essential for systems where zero data loss is mandatory, like financial transactions."

Q: What is cold standby and when is it acceptable?

"Cold standby means the backup is offline — recovery depends on restoring from periodic snapshots or backups. Recovery takes hours and you lose data since the last backup. It is acceptable for dev/staging environments, non-critical applications, or when cost is the primary concern and the business can tolerate extended downtime."

Q: Complete one-liner for interviews.

"Cold standby is offline with periodic backups — recovery takes hours with data loss. Warm standby is a running replica with async replication — recovery takes minutes with possible seconds of data loss. Hot standby is a fully synced replica with synchronous replication — recovery takes seconds with zero data loss. In AWS, RDS Multi-AZ is hot, Read Replicas are warm, and snapshot restores are cold."